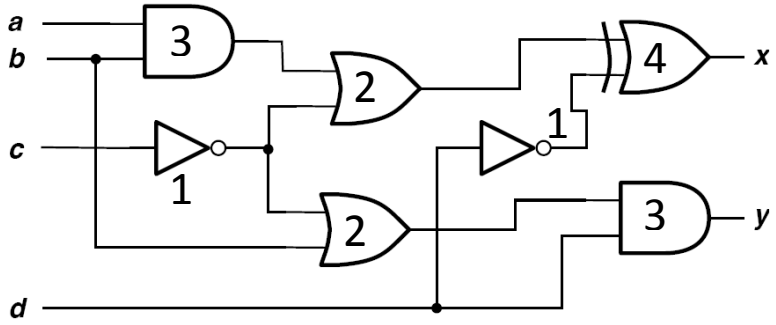
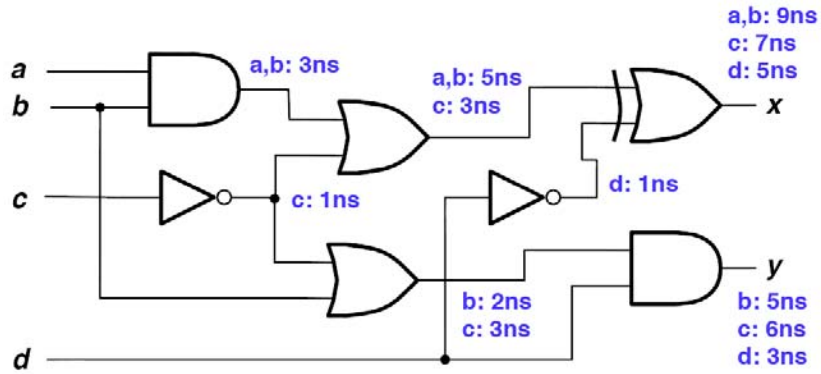
CE2003 Tutorial 5 Solutions

Answer 1

(a) We can trace each input to each output to work out the total delays. Firstly the delay in each gate.



Then the delay through the circuit for each input.

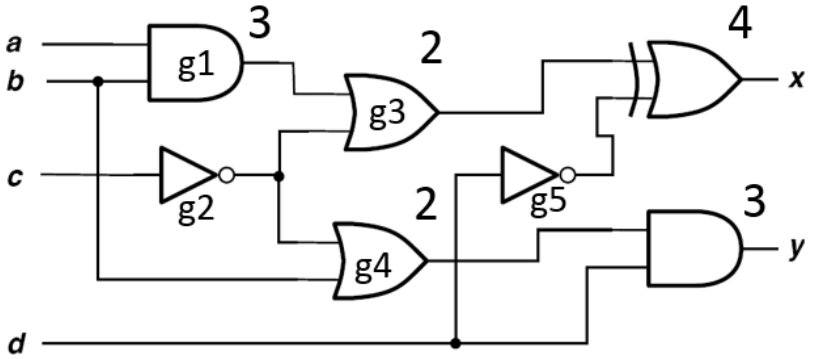


Hence the propagation delays for all paths are:

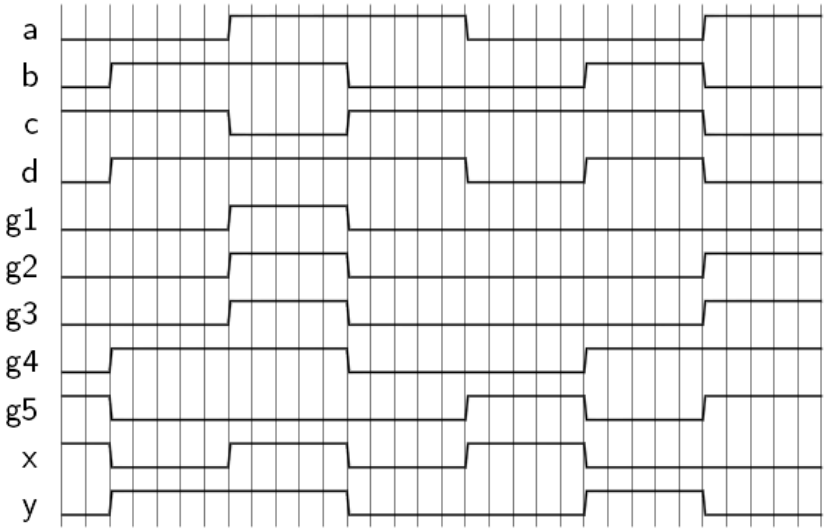
From a or b to x : 9ns
From c to x : 7ns
From d to x : 5ns
From b to y: 5ns
From c to y: 6ns
From d to y: 3ns

(b) Thus the critical path is from a or b to x, through an **and**, **or**, and **xor** gate; a path of 9ns. We could be able to clock this circuit at a maximum frequency of 1/9ns = 111MHz.

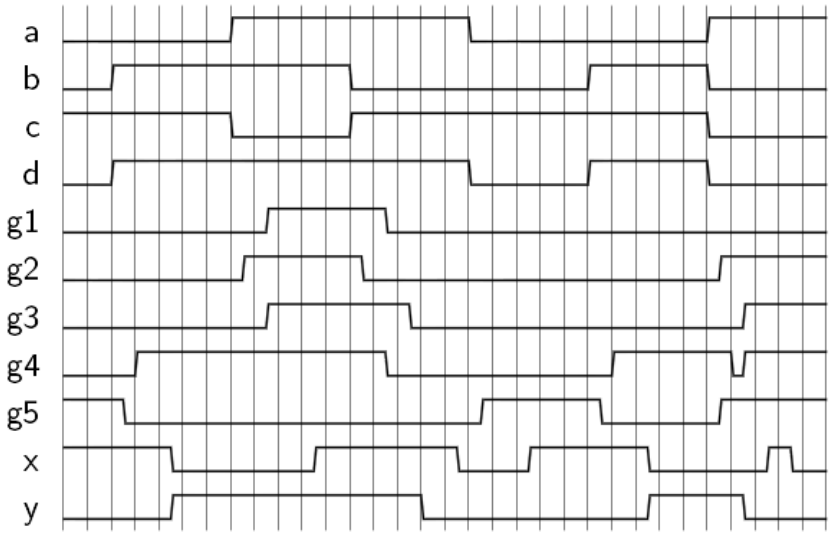
(c) If we label the output of the leftmost **and** gate g1, the leftmost **inverter** g2, the upper middle **or** gate g3, the lower **or** gate g4, the last **inverter** g5, and the outputs as they are.



If there are no delays, we get:



Adding delays, we get



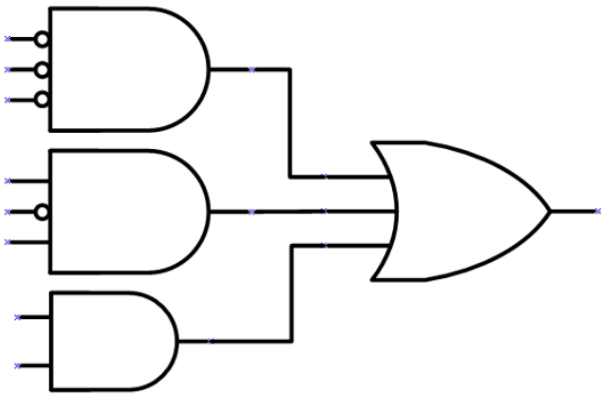
The short low pulse at the end of g4 is a glitch resulting in the slight difference in time between its inputs arriving. This causes the small glitch at the end of the x waveform.

Answer 2

		x[1:0]			
		00	01	11	10
x[3:2]	00	1	0	0	0
	01	0	1	0	0
	11	0	1	1	1
	10	1	0	1	1

		x[1:0]			
		00	01	11	10
x[3:2]	00	1	0	0	0
	01	0	1	0	0
	11	0	1	1	1
	10	1	0	1	1

(~x[2]&~x[1]&~x[0]) | (x[2]&~x[1]&x[0]) | (x[3]&x[1])

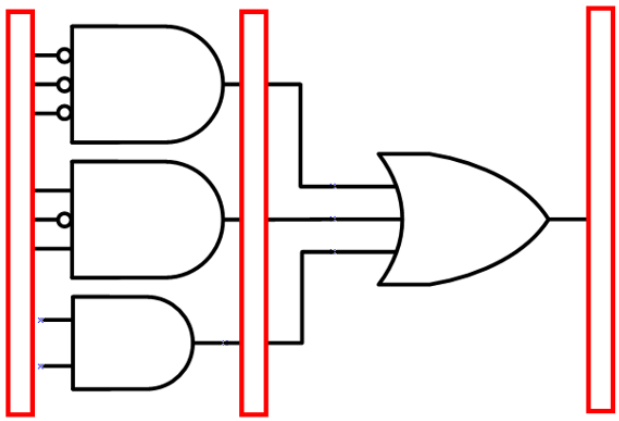


Hence worst case delay is 4 + 4 = 8ns

(b) In an FPGA, this could fit in a 4-input LUT. The propagation delay would be the delay through the LUT, regardless of the function, or whether we simplify it.

(c) We would need to compute $T > t_{cQ} + t_p + t_s$, so $T > 0.6 + 8 + 0.3$, $T > 8.9$ns, so $F_{max} \leq 1/8.9 = 112.3$MHz.

(d)



The longest combinational path is now 4ns, so $T > 0.6 + 4 + 0.3$, $T > 4.9$ns, and $F_{max} = 1/4.9 = 204$MHz.

Answer 3

(a) Longest path is 15ns, so $F_{max} = 1/15 = 66.6$MHz.

(b) Max. frequency will always be the inverse of longest combinational path, so f_c which has a propagation delay of 8ns would limit $F_{max} = 1/8 = 125$MHz.

(c) No register is required between f_b and f_d, since they can both complete in the same time as stage as f_c. So two extra stages are sufficient.

(d)

